

DAR ES SALAAM INSTITUTE OF TECHNOLOGY



DEPARTMENT: COMPUTER DEPARTMENT

MODULE NAME: MOBILE APPLICATION DEVELOPMENT

MODULE CODE: ITT 06214

CLASS: OD23IT

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PROJECT PROPOSAL

PROJECT TITLE: SMART MARINE BOOKING MOBILE APPLICATION

1. Introduction

Marine transportation plays an important role in Tanzania, especially in areas such as Dar es Salaam, Zanzibar, and Mwanza. Many passengers depend on boats for transportation and tourism activities.

However, most existing booking systems are web-based or manual, making it difficult for users who rely on smartphones for accessing services. This project proposes the development of a mobile application that allows users to search, book, and manage boat transport services directly from their mobile devices, creating a dedicated mobile application with improved accessibility and user experience.

2. Problem Statement

Current boat booking methods are not convenient for many users because they require physical visits or depend heavily on websites.

These methods can be time-consuming and difficult to access on mobile devices. Therefore, there is a need for a mobile-based booking system that enables passengers to access boat services anytime and anywhere using their smartphones.

3. Objectives of the Project

The main objective of this project is to develop a mobile application for booking boats in Tanzania.

Specific objectives include:-

1. To develop a subsystem that allows users to register and log into the system.
2. To develop a subsystem that allows users to search available boats and schedules.
3. To develop a subsystem that enables users to book tickets online,
4. To develop a subsystem that enables users to make secure mobile payments
5. To develop a subsystem that provides users ability to receive booking confirmations and notifications.

4. Proposed System Features

The application will include features such as

1. User registration and login
2. Search of boats by destination
3. Viewing of schedules
4. Online booking
5. Payment integration

6. Digital ticket generation
7. Booking history
8. notifications
9. Profile management.

The admin side will allow management of boats, routes, schedules, ticket prices, and customer bookings.

5. Users of the System

The proposed system will mainly be used by passengers and system administrators. Passengers, including travelers, tourists, students, and business people, will use the mobile application to search for available boats, view travel schedules, book tickets, make payments, and receive booking confirmations through their smartphones.

On the other hand, system administrators, such as boat company staff and transport operators, will use the system to manage boats, routes, schedules, ticket prices, and customer bookings. They will also monitor system activities and generate reports to ensure smooth operation of the booking services.

6. Technologies to Be Used

The mobile application will be developed as a native Android application using Java as the programming language. The user interface will be built using XML layouts in Android Studio, with components such as ConstraintLayout, RecyclerView, and CardView to create a clean and responsive design. Data will be stored locally on the device using SQLite through Android's built-in SQLiteOpenHelper class, which satisfies the assignment requirement for local data storage. The application will use the Android LocationManager to access the device GPS sensor for detecting the user's nearest port and providing location-based features. For web API integration, the app will connect to the OpenWeatherMap API to fetch live marine weather data, and to a lightweight REST API built with Node.js and hosted on Render.com to serve route schedules and booking information. HTTP requests will be handled using the Volley library. Google Maps SDK for Android will be used to display route maps and port locations. The primary development tool will be Android Studio, and all source code will be version-controlled using Git and GitHub, with commits from every group member to show individual contributions.

7. Significance of the Project

This project will improve access to marine transport services, reduce long queues at booking offices, save time for passengers, and support digital transformation in Tanzania's transportation sector. The system will also improve customer convenience and contribute to tourism and business activities.

8. Conclusion

The proposed mobile boat booking application aims to provide a modern, reliable, and user-friendly platform for marine transport booking services in Tanzania. The project is relevant, practical, and beneficial to both passengers and transport companies, making it suitable for academic and real-world implementation.